

# Emotional Encryption CTF Write-up

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## Challenge Overview

We intercepted a message from Nemesis that looked like a random stream of transportation/road emojis. The statement “he always starts from an origin point 🚦 and uses a simple 3 digits numeric key” strongly suggests a deterministic cipher rather than noise. The message also included 0 0 , hinting at a coordinate/grid-based origin.

## Initial Analysis

The ciphertext contains a small repeatable emoji alphabet (transport signs, roads, tracks). These emojis belong to the same Unicode “transport/map” region, which is perfect for building a compact numeric encoding.

The core hint is the 3-digit key: 239 . The most natural interpretation is a repeating digit pattern: [2, 3, 9] applied cyclically across the stream (classic poly-shift).

## Decoding Strategy

The goal is to transform each emoji into a small integer, reverse the keyed transformation, and rebuild readable ASCII. The workflow:

- **Step 1:** Tokenize the message into individual emojis.
- **Step 2:** Convert each emoji into a compact integer representation (offset/index).
- **Step 3:** Apply the repeating key digits 2, 3, 9 to de-shift values.
- **Step 4:** Use “path separators” (track/road emojis) to split encoded chunks.
- **Step 5:** Convert the reconstructed byte stream into ASCII to reveal the flag.

## Why the Key Works

A 3-digit key is a dead giveaway for a repeating transform: each symbol is shifted by one digit, then the next, then the next, looping forever. This is lightweight enough to hide content from dumb keyword filters while still being reversible.

## Solver (Reference Implementation)

Below is a clean solver skeleton showing the exact logic used: tokenize  $\rightarrow$  map  $\rightarrow$  de-shift with `[2, 3, 9]`  $\rightarrow$  rebuild plaintext. (Delimiter emojis can be treated as chunk boundaries.)

[illegible]

```
print(bytes(out))
```

## Flag

After applying the key-based inverse transform and reconstructing the ASCII stream, the hidden text resolves to the final flag:

```
IDEH{Emojis_Are_The_New_Hexxx_H4H4}
```

## Key Takeaways

- Emoji streams are often just a compact alphabet for a substitution/shift cipher.
- A “3-digit key” usually means a repeating pattern, not a one-time pad.
- Origin hints ( `0 0` , “start from 🐙”) often imply a structured decoding model (grid/path/index).
- Small “delimiter” symbols (roads/tracks) commonly separate characters or blocks.